

# Text Style Transfer

Jonathan Ruser

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- Encoder-Decoder Architectures
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# Style of Text

“Lend me your ears!”  
(Old English Style)

“Listen  
to me!”

“Listen to me  
y’all!” (Southern  
U.S. Style)

“Please, listen to  
me!” (Polite Style)

# Style Transfer

“Lend me your ears!”  
(Old English Style)



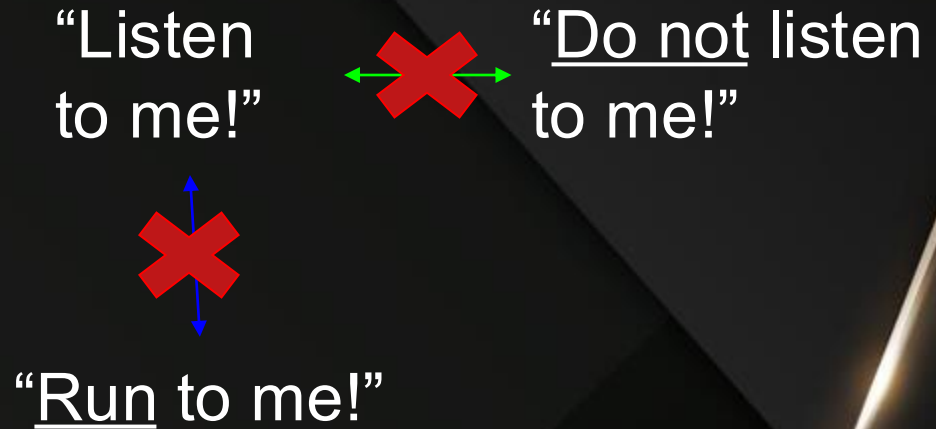
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# Style Vs. Content

- Changing the content changes the meaning
- Some currently explored attributes arguably are content transferring not style transferring



# Uses of Style Transfer

Allowing technical papers to be more accessible to the laymen

“The lesion slowly enlarges, often ulcerates and spread to other skin areas.”



“The sores slowly enlarge and spread to nearby tissue, causing further damage.”

Cao et al. 2020

Adding creativity to text

“She was fast.”



“She was as quick as a cheetah.”

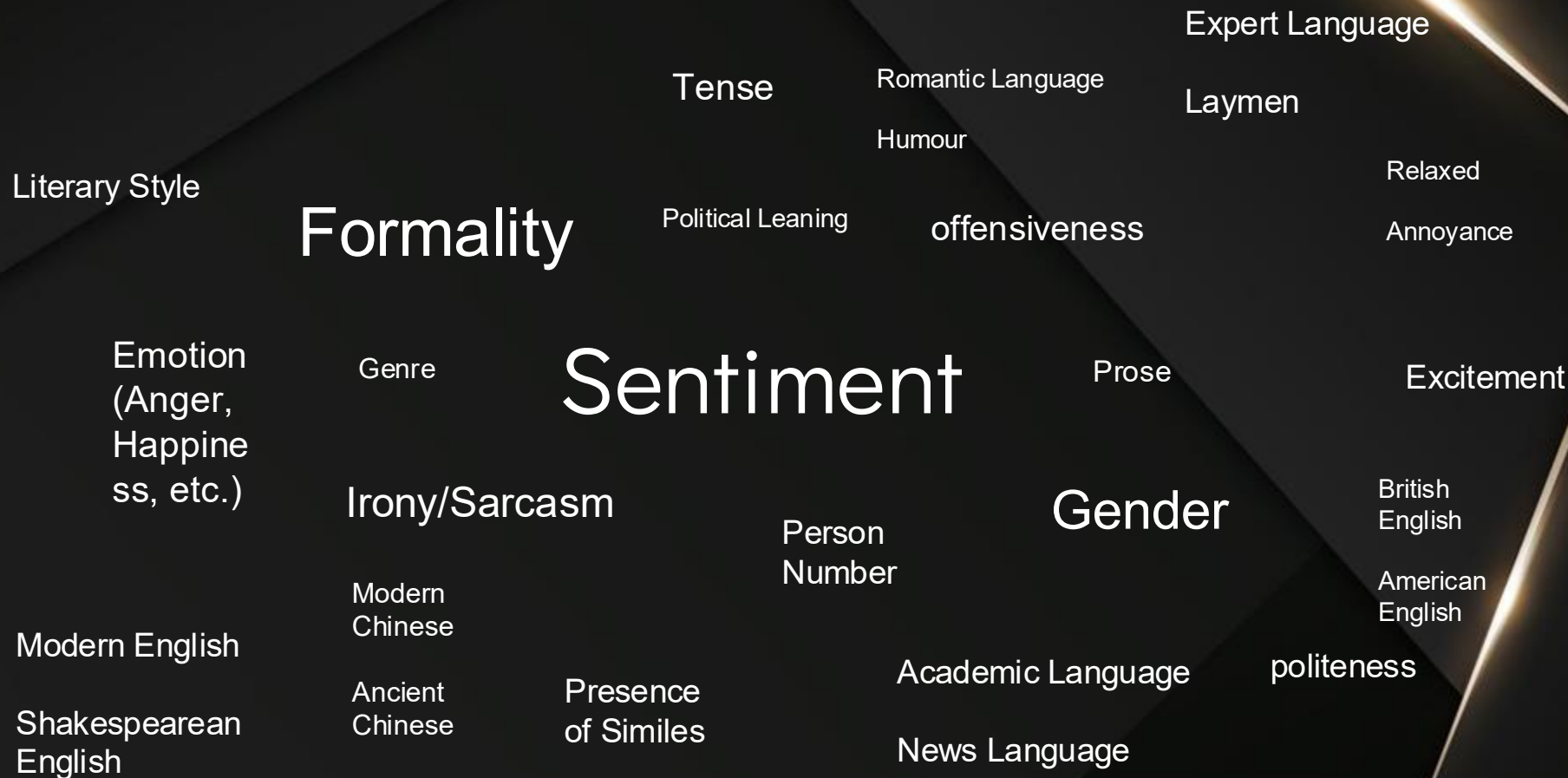
Better understanding of older texts

“And if ye salute your brethren only, what do ye more than others? do not even the publicans so?”



“And if you greet only your own people, what are you doing more than others? Do not even pagans do that?”

# Explored Style Attributes



# Style Attributes

- Each characteristic of a style is an attribute
- Each attribute has multiple values (research often focuses on binary values)
- Style, attribute, value used interchangeably throughout research

| Text               | Attributes |            |             |
|--------------------|------------|------------|-------------|
|                    | Formality  | Politeness | Annoyance   |
| "Let me know ASAP" | informal   | impolite   | not annoyed |



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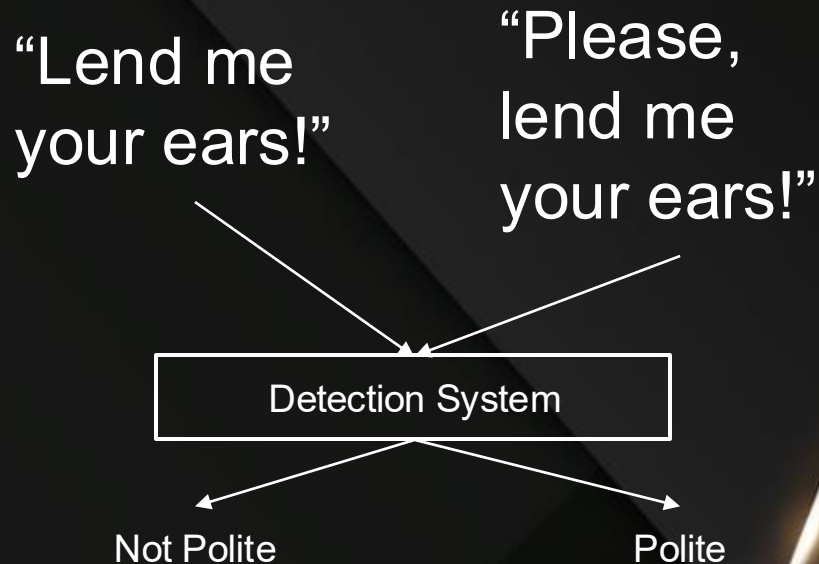
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| "Respond in a timely manner."        | formal     | impolite   | not annoyed |
| "Please get back to me soon."        | informal   | polite     | not annoyed |
| "Still waiting for your response..." | informal   | impolite   | annoyed     |

# Style Transfer vs. Style Detection

- Style detection (attribute detection) first logical step
  - Detection algorithms sometimes used in transfer
- Style transfer follows well explored detection areas
  - Datasets readily available



# Style Transfer Worksheet

# Encoder-Decoder Characteristics

- From surveying the large amount of papers in style transfer, we note the main characteristics which differ between style transfer systems
- Though the base architecture is similar, the choice of characteristics causes models to differ:
  - Loss Functions
  - Disentanglement
  - Data Alignment
  - Multi-attribute or Single Attribute
  - Input into generator
  - Input of Style
  - Discriminators

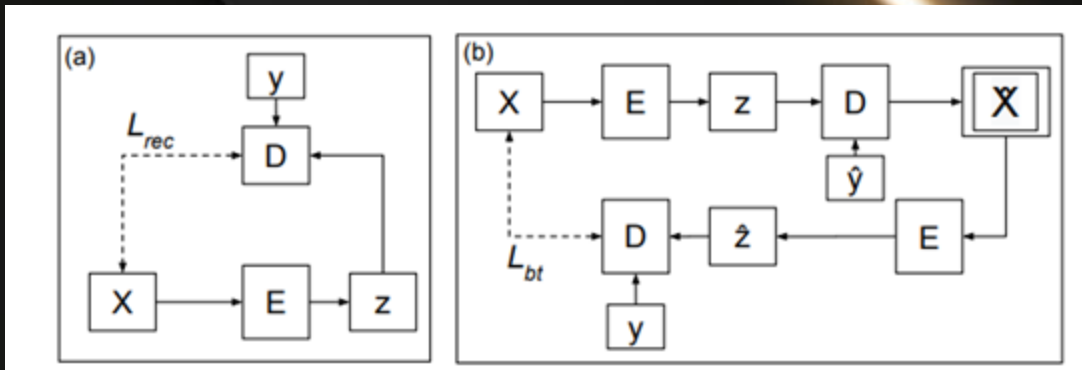
# Loss Functions

- (a) Reconstruction Loss

$$L_{rec} = -\log P(X|z, y)$$

- (b) Back-translation Loss

$$L_{bt} = -\log P(X|\hat{z}, y)$$

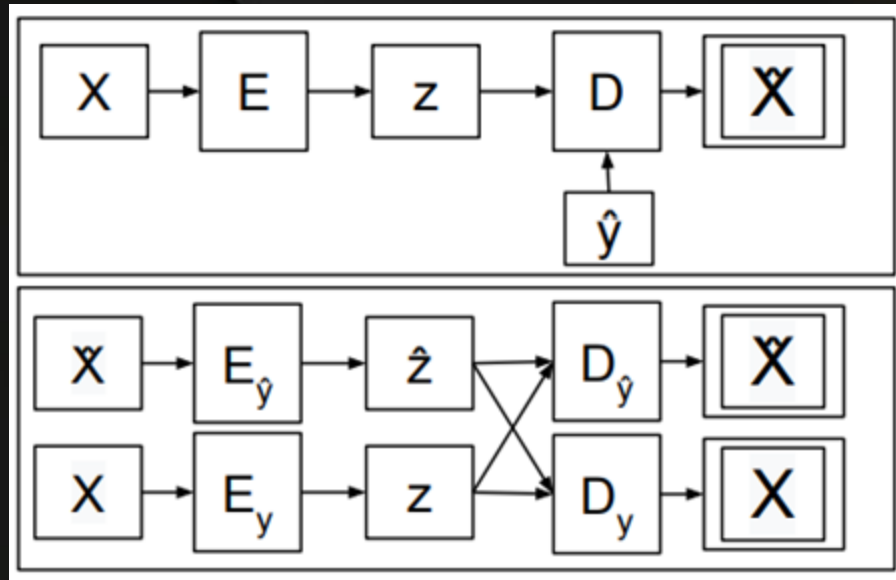


Others:

- Classification Loss
- Adversarial Loss
- Fluency
- Content Preservation
- Reference

# Input of Style

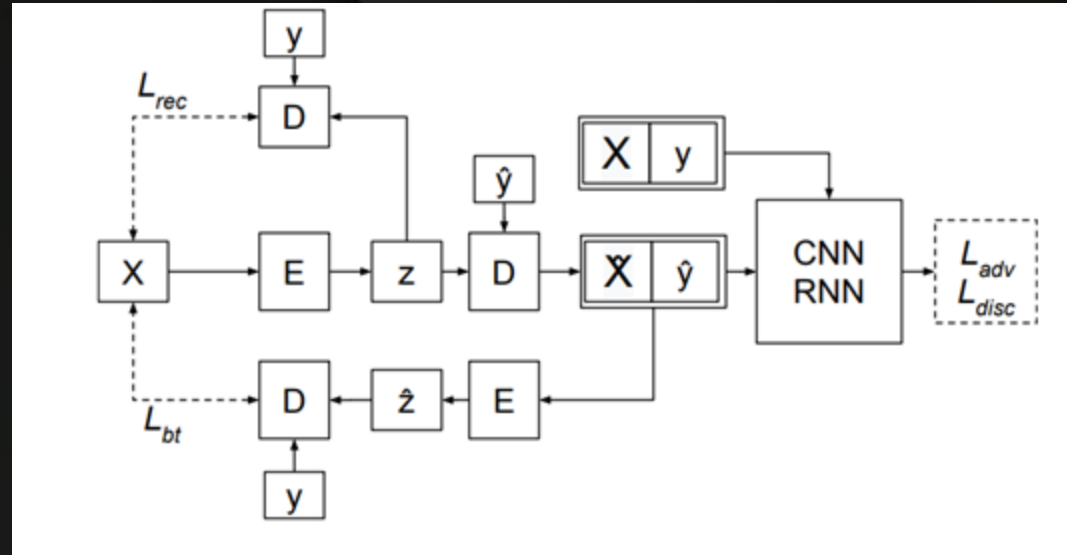
- How is target style introduced?
- Added to encoded input
- Train decoders on target styles





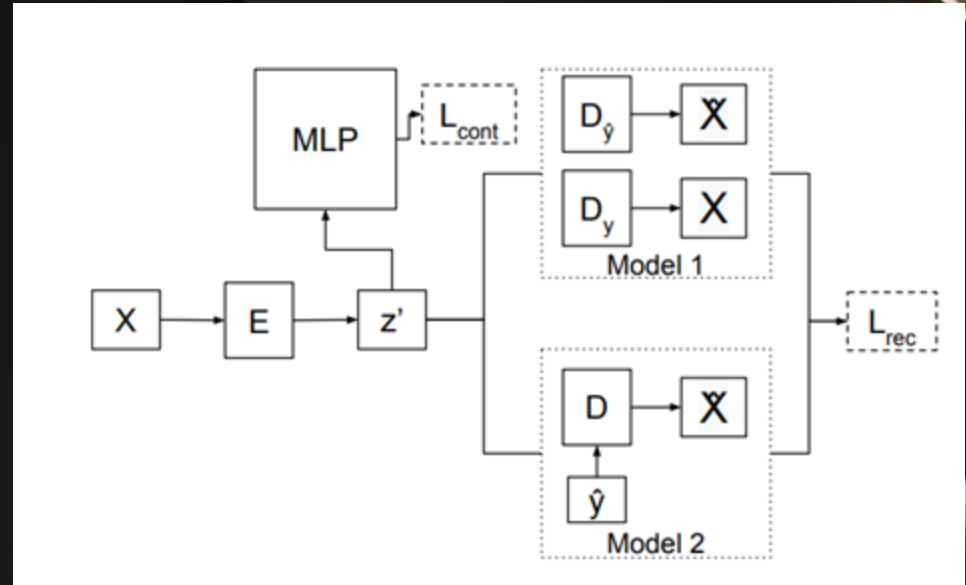
# [Lai et al., 2019]

- Focus on sentiment (Yelp, Amazon), tense (Yelp), as well as types of products (Amazon).
- GRU based encoder-decoder
- Reconstruction, backtranslation, classification, adversarial loss
- No disentanglement, nonparallel data, input is not modified
- Style embedding is concatenated to context vector
- Use both CNN and RNN



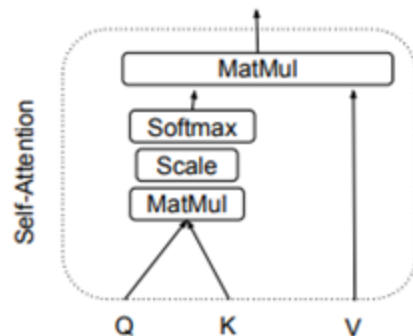
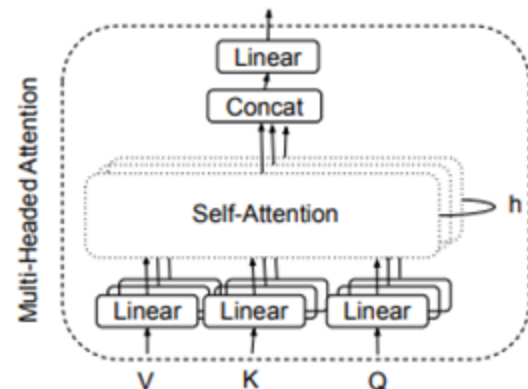
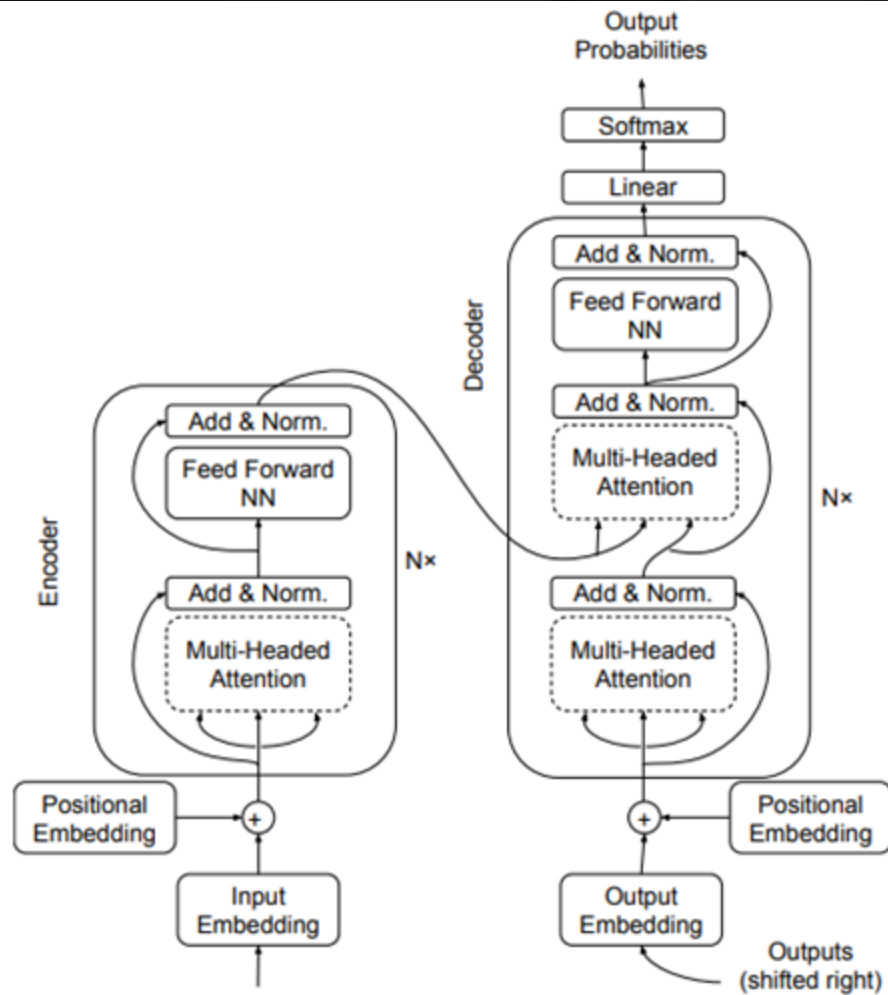
# [Fu et al., 2017]

- Focus on sentiment transfer and paper titles (academic to news)
- Reconstruction loss, content loss (for disentanglement)
- Non-parallel data, single attribute, input not modified
- Input of style depends on model
- Multi Layer Perceptron (MLP) used for disentanglement

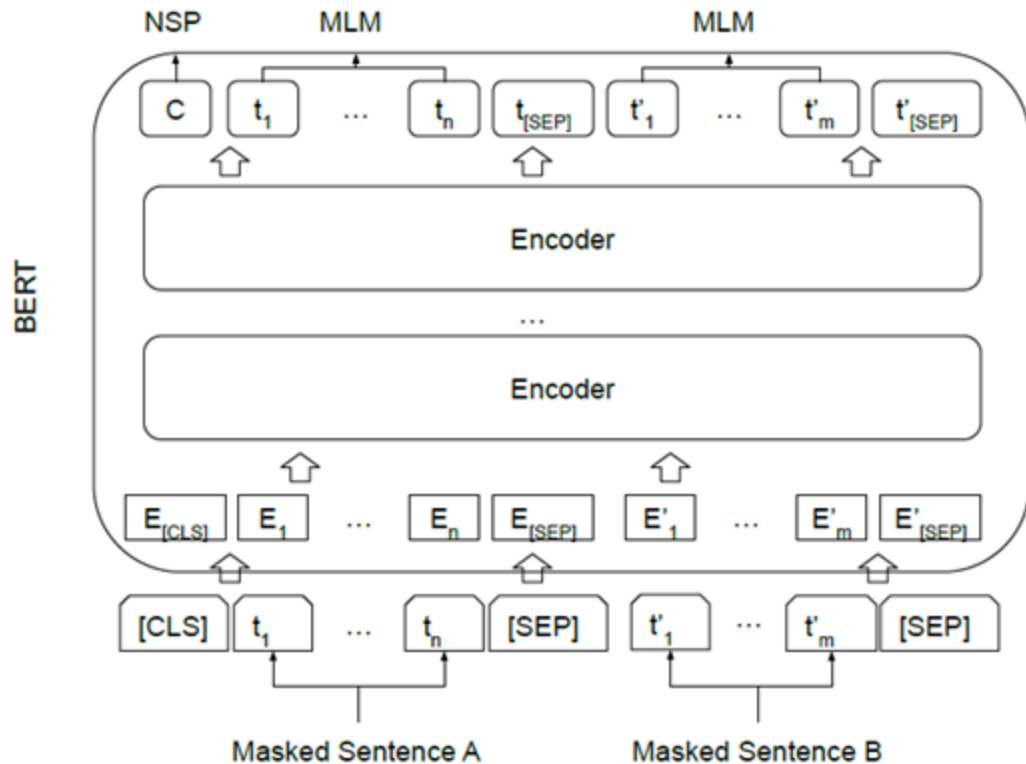


# Transformers for Style Transfer

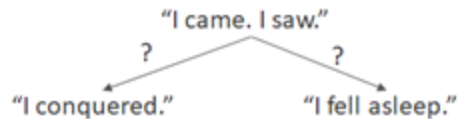
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- **Transformer Architectures**
  - Characteristics
  - Example
- Measurements



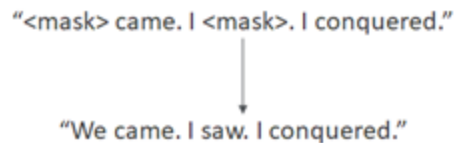
# BERT



## Next Sentence Prediction (NSP)



## Masked Language Modeling (MLM)



# Transformers in Style Transfer

## “Mask and Infill”



## “Delete and generate”



# Transformer Characteristics

- Mask/Deletion Approach
- How style is determined
- Infiller/Generator Architecture
- Input versus output length
- External Classifiers
- Loss Functions

# Mask/Deletion Approach

How are style indicating tokens masked or deleted?

- Attention of classification systems
- Frequency ratio
- Machine learning system to recognize where to add mask
- Disagreement between BERT models trained on source and target style

## Attention of classification systems

“This restaurant was fantastic! The beds were clean, the room service was perfect!”

## Disagreement between BERT models trained on source and target style

Bert<sub>Positive</sub>

“This restaurant was fantastic! The beds were clean, the room service was perfect!”

Bert<sub>Negative</sub>

“This restaurant was fantastic! The beds were clean, the room service was perfect!”



# Input vs output length

- Is the output length limited by the input length?

“Once upon a midnight <mask>, while I  
pondered, <mask>,  
Over many a <mask> and curious volume of  
<mask>—”

“Once upon a midnight dreary, while I  
pondered, weary,  
Over many a quaint and curious volume of  
lore—”

“Once upon a midnight dreary, while I  
pondered, weak and weary,  
Over many a quaint and curious volume of  
forgotten lore—”

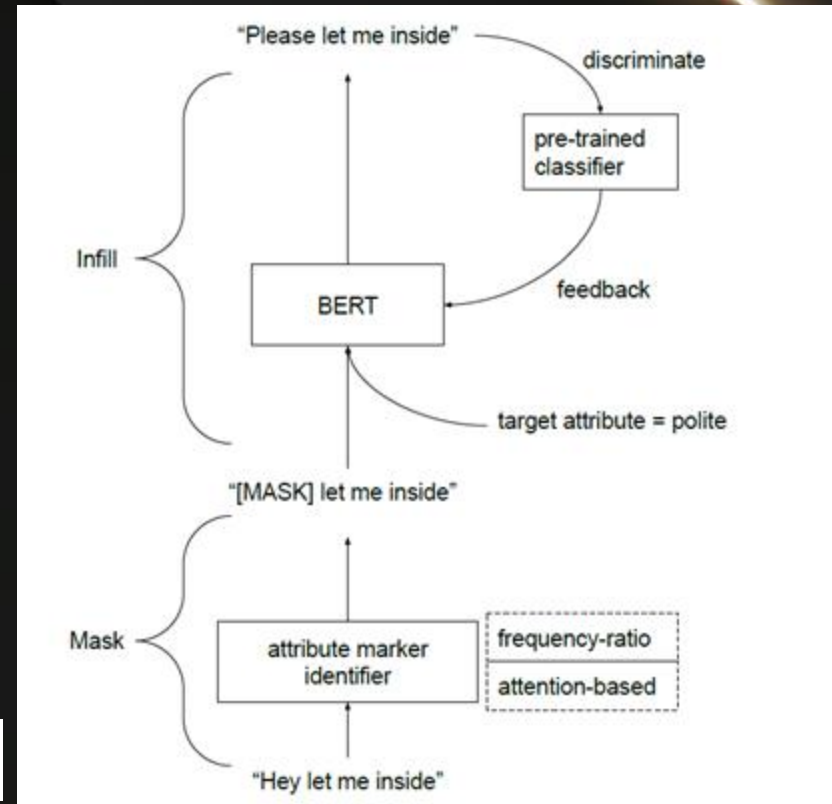
# [Wu et al., 2019a]

- Focus on sentiment
- Attention/Frequency ratio combination for masking
- Substitute attribute embeddings for segment embeddings
- BERT used for infiller
- CNN used for sentiment feedback

$$\mathcal{L}_{rec} = - \sum_{a \in \mathcal{A}, t_i \in M} \log p(t_i | \bar{S}, a)$$

$$\mathcal{L}_{acc} = -\log p(\hat{a} | \hat{S})$$

$$\min_{\theta} L = L_{rec} + \mu L_{acc}$$



# Measurements

- Overview
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- **Measurements**

# Measuring Effectiveness

## Automatic Evaluation

### Content Preservation

- BLEU, P<sub>INC</sub>, TER<sub>p</sub>, METEOR, ROUGE

- Accuracy (Transfer Strength)

- $$overall = \frac{BLEU \times Accuracy}{BLEU + Accuracy}$$

- Fluency

## Human Evaluation

- Rating text for fluency, accuracy, content preservation on a scale (e.g. 1-5)
- Comparing outputs of 2 different systems
- Rank multiple system outputs at once



**Questions?**